



Lab 4:
Packet Tracer Simulation
(Exploration of ARP and Switch Table Communications)

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PART 1: Review the Topology

Step1

[5m]

- a)
- ```

RTA>enable
RTA#show arp
Protocol Address Age (min) Hardware Addr Type Interface
Internet 192.168.11.1 - 0002.4A00.0E91 ARPA FastEthernet1/0
Internet 192.168.33.1 - 000C.CF0C.593A ARPA FastEthernet0/0
RTA#

```
- b)
- ```

RTB>enable
RTB#show arp
Protocol Address      Age (min) Hardware Addr  Type   Interface
Internet 192.168.4.1        -    0001.977A.B614  ARPA   FastEthernet0/0
RTB#
    
```
- c) **SWA**
- ```

SWA>enable
SWA#show arp

SWA#show mac-address-table
 Mac Address Table

Vlan Mac Address Type Ports
---- -
 1 0002.4a00.0e91 DYNAMIC Fa0/1
 1 000c.8546.7d85 DYNAMIC Fa1/1
SWA#

```
- SWB**
- ```

SWB>enable
SWB#show arp

SWB#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----    -
      1    000c.cf0c.593a    DYNAMIC   Fa0/1
SWB#
    
```
- SWC**
- ```

SWC>enable
SWC#show arp

SWC#show mac-address-table
 Mac Address Table

Vlan Mac Address Type Ports
---- -
 1 0001.977a.b614 DYNAMIC Fa0/1
SWC#

```

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p>d) <b><u>PCA</u></b></p> <pre>Cisco Packet Tracer PC Command Line 1.0 C:\&gt;arp -a No ARP Entries Found C:\&gt;</pre> <p><b><u>PCB</u></b></p> <pre>Cisco Packet Tracer PC Command Line 1.0 C:\&gt;arp -a No ARP Entries Found C:\&gt;</pre> <p><b><u>PCC</u></b></p> <pre>Cisco Packet Tracer PC Command Line 1.0 C:\&gt;arp -a No ARP Entries Found C:\&gt;</pre> <p><b><u>PCD</u></b></p> <pre>Cisco Packet Tracer PC Command Line 1.0 C:\&gt;arp -a No ARP Entries Found C:\&gt;</pre> <p><b><u>PCE</u></b></p> <pre>Cisco Packet Tracer PC Command Line 1.0 C:\&gt;arp -a No ARP Entries Found C:\&gt;</pre> |
|  | <p>e) No entry for all PCs because all the PCs didn't receive any ARP request.<br/>There will be no entry in the PC ARP table.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

## PART 2: Generate Network Traffic

|                   |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Step<br>1<br>[3m] | c) | <pre> Cisco Packet Tracer PC Command Line 1.0 C:\&gt;arp -a No ARP Entries Found C:\&gt;ping 192.168.33.22  Pinging 192.168.33.22 with 32 bytes of data:  Reply from 192.168.33.22: bytes=32 time=5ms TTL=128 Reply from 192.168.33.22: bytes=32 time&lt;1ms TTL=128 Reply from 192.168.33.22: bytes=32 time&lt;1ms TTL=128 Reply from 192.168.33.22: bytes=32 time&lt;1ms TTL=128  Ping statistics for 192.168.33.22:     Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),     Approximate round trip times in milli-seconds:         Minimum = 0ms, Maximum = 5ms, Average = 1ms  C:\&gt;arp -a     Internet Address      Physical Address      Type     192.168.33.22         0060.47ea.a746       dynamic </pre> |
|                   | d) | <pre> Cisco Packet Tracer PC Command Line 1.0 C:\&gt;arp -a No ARP Entries Found C:\&gt;arp -a     Internet Address      Physical Address      Type     192.168.33.33         0002.1755.9a06       dynamic </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                   | e) | <p><b><u>PCC</u></b></p> <pre> Cisco Packet Tracer PC Command Line 1.0 C:\&gt;arp -a No ARP Entries Found C:\&gt;arp -a No ARP Entries Found </pre> <p><b><u>PCD</u></b></p> <pre> Cisco Packet Tracer PC Command Line 1.0 C:\&gt;arp -a No ARP Entries Found C:\&gt;arp -a No ARP Entries Found </pre> <p><b><u>PCE</u></b></p> <pre> Cisco Packet Tracer PC Command Line 1.0 C:\&gt;arp -a No ARP Entries Found C:\&gt;arp -a No ARP Entries Found </pre>                                                                                                                                                                                                                                                              |

|                            |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |
|----------------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <p>Step<br/>2<br/>[7m]</p> | <p>b)</p> | <pre>C:\&gt;ping 192.168.33.22  Pinging 192.168.33.22 with 32 bytes of data:  Request timed out. Reply from 192.168.33.22: bytes=32 time&lt;1ms TTL=127 Reply from 192.168.33.22: bytes=32 time&lt;1ms TTL=127 Reply from 192.168.33.22: bytes=32 time&lt;1ms TTL=127  Ping statistics for 192.168.33.22:     Packets: Sent = 4, Received = 3, Lost = 1 (25% loss), Approximate round trip times in milli-seconds:     Minimum = 0ms, Maximum = 0ms, Average = 0ms  C:\&gt;arp -a     Internet Address      Physical Address        Type 192.168.11.1             0002.4a00.0e91         dynamic</pre>                                                                                                   |  |
|                            | <p>c)</p> | <pre>C:\&gt;ping 192.168.11.112  Pinging 192.168.11.112 with 32 bytes of data:  Reply from 192.168.11.112: bytes=32 time&lt;1ms TTL=128 Reply from 192.168.11.112: bytes=32 time&lt;1ms TTL=128 Reply from 192.168.11.112: bytes=32 time&lt;1ms TTL=128 Reply from 192.168.11.112: bytes=32 time&lt;1ms TTL=128  Ping statistics for 192.168.11.112:     Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds:     Minimum = 0ms, Maximum = 0ms, Average = 0ms  C:\&gt;arp -a     Internet Address      Physical Address        Type 192.168.11.1             0002.4a00.0e91         dynamic 192.168.11.112           0001.6462.0278         dynamic</pre> |  |
|                            | <p>d)</p> | <pre>C:\&gt;ping 192.168.4.110  Pinging 192.168.4.110 with 32 bytes of data:  Request timed out. Reply from 192.168.4.110: bytes=32 time=10ms TTL=126 Reply from 192.168.4.110: bytes=32 time=10ms TTL=126 Reply from 192.168.4.110: bytes=32 time=9ms TTL=126  Ping statistics for 192.168.4.110:     Packets: Sent = 4, Received = 3, Lost = 1 (25% loss), Approximate round trip times in milli-seconds:     Minimum = 9ms, Maximum = 10ms, Average = 9ms  C:\&gt;arp -a     Internet Address      Physical Address        Type 192.168.11.1             0002.4a00.0e91         dynamic 192.168.11.112           0001.6462.0278         dynamic</pre>                                                 |  |

|                |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |  |  |  |  |
|----------------|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|
|                | e) | The ARP entries are revealed in the ARP table only after PCC pings PCB, PCD, and PCE. No ARP entries are found before these ping activities. The ARP table displays entries once PCB, PCD, and PCE receive the ARP request from PCC and subsequently receive pings from PCB, PCD, and PCE.                                                                                                                                                                               |  |  |  |  |  |
|                | f) | <pre>RTA&gt;enable RTA#show arp Protocol  Address                Age (min)  Hardware Addr  Type   Interface Internet  192.168.11.1            -          0002.4A00.0E91  ARPA    FastEthernet1/0 Internet  192.168.11.11          5          00D0.D39A.C0D9  ARPA    FastEthernet1/0 Internet  192.168.33.1           -          000C.CF0C.593A  ARPA    FastEthernet0/0 Internet  192.168.33.22          5          0060.47EA.A746  ARPA    FastEthernet0/0 RTA# </pre> |  |  |  |  |  |
|                | g) | <pre>RTB&gt;enable RTB#show arp Protocol  Address                Age (min)  Hardware Addr  Type   Interface Internet  192.168.4.1            -          0001.977A.B614  ARPA    FastEthernet0/0 Internet  192.168.4.110          2          0060.702D.7C08  ARPA    FastEthernet0/0 RTB# </pre>                                                                                                                                                                          |  |  |  |  |  |
| Step 3<br>[7m] | a) | <pre>SWA&gt;enable SWA#show arp  SWA#show mac-address-table       Mac Address Table ----- Vlan    Mac Address                Type           Ports ----    - 1       0001.6462.0278             DYNAMIC        Fa1/1 1       0002.4a00.0e91             DYNAMIC        Fa0/1 1       000c.8546.7d85             DYNAMIC        Fa1/1 1       00d0.d39a.c0d9             DYNAMIC        Fa2/1 SWA# </pre>                                                                  |  |  |  |  |  |
|                | b) | <pre>SWB&gt;enable SWB#show arp  SWB#show mac-address-table       Mac Address Table ----- Vlan    Mac Address                Type           Ports ----    - 1       000c.cf0c.593a             DYNAMIC        Fa0/1 SWB# </pre>                                                                                                                                                                                                                                          |  |  |  |  |  |

|    |                                                                                                                                                                                                                                         |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| c) | <pre>SWC&gt;enable SWC#show arp  SWC#show mac-address-table       Mac Address Table ----- Vlan    Mac Address      Type    Ports ----    - 1        0001.977a.b614    DYNAMIC Fa0/1 1        0060.702d.7c08    DYNAMIC Fa1/1 SWC#</pre> |
| d) | Yes                                                                                                                                                                                                                                     |
| e) | Switches utilize an ARP table to store the IP addresses and corresponding MAC addresses of network devices. This table serves the purpose of identifying the destination MAC addresses for communication with network nodes.            |
| f) | To view MAC address table entries on a switch, the term "DYNAMIC" specifically pertains to displaying MAC addresses that have been learned by the switch during its operation.                                                          |

### PART 3: Attach Wireless Lab Results

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [1m] | a) Filename: LING YU QIAN_Lab4.pka                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| [3m] | <p>c)</p> <p>The screenshot displays a Packet Tracer workspace with a network diagram and a configuration window. The network diagram shows a central switch S1 connected to a wireless router WRS2, which is connected to PC3. S1 has subinterfaces for VLAN 10 (PC1) and VLAN 20 (PC2). WRS2 has a wireless LAN interface connected to PC3. The configuration window shows steps for setting up wireless access on PC3, including naming the profile, setting the SSID to WRS_LAN, and configuring security with WPA2-Personal and the passphrase cisco123.</p> |

